

Niveau Moyen — Corrigé

Corrigé 1

$\int \ln x \, dx$. On pose $U = \ln x$, $V' = 1$, donc $U' = \frac{1}{x}$, $V = x$.

$$\int \ln x \, dx = x \ln x - \int x \cdot \frac{1}{x} \, dx = x \ln x - \int 1 \, dx = \boxed{x \ln x - x + C.}$$

Vérif. $[x \ln x - x]' = \ln x + 1 - 1 = \ln x$. ✓

Corrigé 2

$\int x \ln x \, dx$. Règle L.I.A.T.E. : on dérive le $\ln$. $U = \ln x$, $V' = x$, donc $U' = \frac{1}{x}$, $V = \frac{x^2}{2}$.

$$\int x \ln x \, dx = \frac{x^2}{2} \ln x - \int \frac{x^2}{2} \cdot \frac{1}{x} \, dx = \frac{x^2}{2} \ln x - \frac{1}{2} \int x \, dx = \boxed{\frac{x^2}{2} \ln x - \frac{x^2}{4} + C.}$$

Vérif. $[\frac{x^2}{2} \ln x - \frac{x^2}{4}]' = x \ln x + \frac{x^2}{2} \cdot \frac{1}{x} - \frac{x}{2} = x \ln x + \frac{x}{2} - \frac{x}{2} = x \ln x$. ✓

Corrigé 3

$\int \arctan x \, dx$. On pose $U = \arctan x$, $V' = 1$, donc $U' = \frac{1}{1+x^2}$, $V = x$.

$$\int \arctan x \, dx = x \arctan x - \int \frac{x}{1+x^2} \, dx.$$

Pour la dernière intégrale, forme dW/W avec $W = 1+x^2$, $dW = 2x \, dx$: $\int \frac{x \, dx}{1+x^2} = \frac{1}{2} \ln(1+x^2)$. D'où

$$\boxed{\int \arctan x \, dx = x \arctan x - \frac{1}{2} \ln(1+x^2) + C.}$$

Vérif. $[x \arctan x - \frac{1}{2} \ln(1+x^2)]' = \arctan x + \frac{x}{1+x^2} - \frac{1}{2} \cdot \frac{2x}{1+x^2} = \arctan x$. ✓

Corrigé 4

$\int x e^{2x} \, dx$. $U = x$, $V' = e^{2x}$, donc $U' = 1$, $V = \frac{1}{2} e^{2x}$.

$$\int x e^{2x} \, dx = x \cdot \frac{1}{2} e^{2x} - \int \frac{1}{2} e^{2x} \, dx = \frac{x}{2} e^{2x} - \frac{1}{2} \cdot \frac{1}{2} e^{2x} + C = \boxed{\frac{2x-1}{4} e^{2x} + C.}$$

Vérif. $[\frac{x}{2} e^{2x} - \frac{1}{4} e^{2x}]' = \frac{1}{2} e^{2x} + \frac{x}{2} \cdot 2e^{2x} - \frac{1}{4} \cdot 2e^{2x} = \frac{1}{2} e^{2x} + x e^{2x} - \frac{1}{2} e^{2x} = x e^{2x}$. ✓

Corrigé 5

$\int x^2 e^x \, dx$. **1^{re} IPP** : $U = x^2$, $V' = e^x$ ($U' = 2x$, $V = e^x$) :

$$\int x^2 e^x \, dx = x^2 e^x - \int 2x e^x \, dx = x^2 e^x - 2 \int x e^x \, dx.$$

2^e IPP (déjà connue) : $\int x e^x \, dx = (x - 1)e^x + C$. Donc

$$\int x^2 e^x \, dx = x^2 e^x - 2(x - 1)e^x + C = \boxed{(x^2 - 2x + 2) e^x + C}.$$

Vérif. $[(x^2 - 2x + 2)e^x]' = (2x - 2)e^x + (x^2 - 2x + 2)e^x = (x^2)e^x = x^2 e^x$. ✓